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Professor Rao

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Contribution Journal

[Min Ma Kha]

I uploaded the Module 2 notebook to Google Colab and shared with my team members, executed all cells with Runtime → Run all, and verified that each cell produced correct output before moving on — confirming the library imports, the 150-row DataFrame with four features and three classes, and both required visualizations.

I completed the Analysis and Reflection markdown cell, filling in the dataset overview (150 samples, 4 features, 3 classes) and writing three findings drawn from the scatter plot and the group means table rather than from the lab instructions.

I interpreted the group means table to explain why the sepal scatter plot leaves versicolor and virginica overlapping: petal length varies sharply across species (1.46 → 4.26 → 5.55 cm) while sepal width stays nearly flat (3.43 → 2.77 → 2.97 cm).

I extended the notebook beyond the 16 provided cells by adding a Further Investigation section — a markdown cell stating the hypothesis and a code cell plotting petal length vs. petal width — which confirmed that petal measurements separate all three species with visible gaps.

[Chris Roy]

For this assignment, I had worked on the reflective journal and documented from my perspective of the struggle and challenge I had seen my groupmates endure secondhand. I had provided my group with documents and practice from other classes and labs to give them a headstart to review the information provided in the google collab.

[Camille Casas]

For further investigation, I focused on examining which features provide the greatest value for accurate species classification. I identified petal length and width as particularly significant for distinguishing *Iris versicolor* from *Iris virginica* and considered using a linear decision-boundary model to establish baseline accuracy. This investigation would provide a stronger foundation for evaluating feature selection and improving future classification performance.

[Rigoberto Padilla]

Assumed absolute ownership over the programming execution, mathematical vector operations, and visual canvas debugging. I successfully coded the mathematical calculations using NumPy vectors to determine the dataset's mean and standard deviation metrics. I engineered the categorical bar chart using Matplotlib to visually confirm the balance of our target features. I

resolved critical runtime `NameError` exceptions by restructuring the code execution path from the top-level cells. I finalized our analytical observations and reflection segments within the production-ready Markdown notebook cells.

[Samantha Mireles]

Worked with the Iris dataset and practiced using pandas DataFrames to view and organize data. Helped troubleshoot and check notebook cells to make sure the code ran successfully without errors. Reviewed the data preprocessing portion and helped understand why preparing data important before machine learning